Effect of Flame Structure on the Flame Transfer Function in a Premixed Gas Turbine Combustor

Daesik Kim
Jong Guen Lee[1]
Bryan D. Quay
Domenic A. Santavicca

Department of Mechanical and Nuclear Engineering,
Penn State University,
University Park, PA 16802

Kwanwoo Kim
Shiva Srinivasan

GE Energy,
Greenville, SC 29615

The flame transfer function in a premixed gas turbine combustor is experimentally determined. The fuel (natural gas) is premixed with air upstream of a choked inlet to the combustor. Therefore, the input to the flame transfer function is the imposed velocity fluctuations of the fuel/air mixture without equivalence ratio fluctuations. The inlet-velocity fluctuations are achieved by a variable-speed siren over the forcing frequency of 75–280 Hz and measured using a hot-wire anemometer at the inlet to the combustor. The output function (heat release) is determined using chemiluminescence measurement from the whole flame. Flame images are recorded to understand how the flame structure plays a role in the global heat release response of flame to the inlet-velocity perturbation. The results show that the gain and phase of the flame transfer function depend on flame structure as well as the frequency and magnitude of inlet-velocity modulation and can be generalized in terms of the relative length scale of flame to convection length scale of inlet-velocity perturbation, which is represented by a Strouhal number. Nonlinear flame response is characterized by a periodic vortex shedding from shear layer, and the nonlinearity occurs at lower magnitude of inlet-velocity fluctuation as the modulation frequency increases. However, for a given modulation frequency, the flame structure does not affect the magnitude of inlet-velocity fluctuation at which the nonlinear flame response starts to appear. [DOI: 10.1115/1.3124664]

Keywords: combustion dynamics, gas turbine, combustion, flame transfer function, flame structure

1 Introduction

Lean premixed gas turbine combustors were successful in meeting current NO_x emissions regulations. However, these have been found to be susceptible to combustion instability, which is intimately related to the feedback coupling between unsteady heat release and acoustic pressure field [1,2].

A quantification of the heat release response of a flame to the unsteady inlet velocity and/or equivalence ratio, which are caused by the pressure waves in a combustor, is critical information in the theoretical description of combustion-induced oscillations. The flame transfer function describes its dynamic heat release response to inlet flow perturbations as a function of frequency (f) and with the inlet velocity as the input function it can be defined as

$$T(f) = \frac{Q'(f)/Q_{\text{mean}}}{V'(f)/V_{\text{mean}}}$$

where Q_{mean} is the temporal mean of heat release rate, V_{mean} is the mean bulk velocity of the mixture entering the combustor, and Q' and V' are their corresponding fluctuating components.

Flame transfer function may be obtained from an analytical description of flame response [3,4], from a computational fluid dynamics (CFD)-based study [5,6], and from an experiment [7–10]. However, both analytic and CFD-based approaches are limited in capturing the highly complex interaction between flow field and heat release precisely, and the experimental determination of the flame transfer function is still preferred. The measured flame transfer function can be used directly in the thermoacoustic network model where a gas turbine combustion system is modeled as a network of its elements such as air supply, plenum, premix duct, fuel injection, flame, combustor, etc. [11,12]. Both analytical and experimental works show that the gain of transfer function decreases as the perturbation frequency increases. And, for a given perturbation frequency, it is constant up to a certain level of perturbation but starts to decrease as the level increases further. The former and the latter behavior of flame response are often referred to as linear and nonlinear responses, respectively, and are known to be important to understand mechanism of combustion instability [13,14] in that linear response determines the frequency and the growth rate of combustion instability and the nonlinear process controls the amplitude of instability.

Analytical predictions based on a kinematic analysis of the flame front evolution show that the flame transfer function depends on flame angle [15] and flame length [14] in addition to the basic shape of undisturbed flame (i.e., conical, inverted wedge, V-flame, etc.). For example, the gain of transfer function is found not to be sensitive to the flame angle for a conical flame but quite sensitive for small flame-angle V-flame [15] (i.e., the gain increases as the flame angle decreases). Similarly, the ratio of the flame length to the width is found to affect the magnitude of velocity at which transfer function nonlinearities become significant [14]. However, there are little data available on detailed measurements of flame transfer function in highly turbulent, practical combustor. It will be interesting to see if the relatively simple kinematic analysis of flame dynamics can predict how flame responds to inlet-velocity distribution under practical operating conditions. Also, most flame transfer function measurements were made from the global heat release measurement where it is very difficult to understand why flame behaves in a certain way for given operating conditions.

The objective of this study is to experimentally determine the transfer function of turbulent premixed flames with main emphasis on the effect of flame structure on how the flames respond to inlet-velocity perturbation. Also, the nonlinear response of flame will be characterized, and controlling parameters of the nonlinearity will be identified.

[1]Corresponding author.
Manuscript received April 1, 2008; final manuscript received May 9, 2008; published online October 16, 2009. Review conducted by Dilip R. Ballal.

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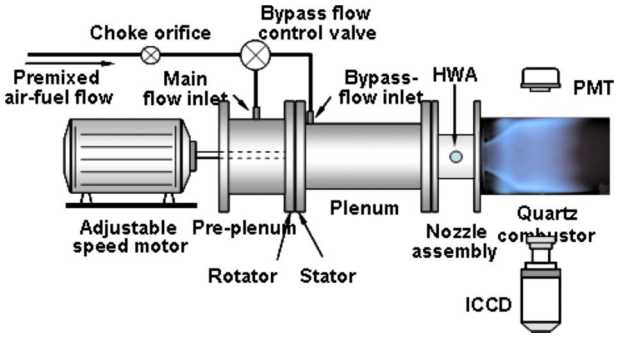

Fig. 1 Schematic diagram of experimental setup

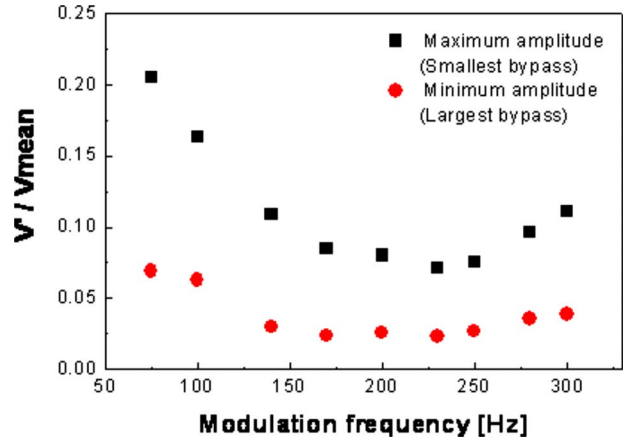

Fig. 3 Maximum and minimum amplitudes of velocity fluctuations at each modulation frequency ($V_{mean,nozzle}$=30 m/s)

2 Experimental Approach

The experimental setup consists of an optically-accessible combustor, a nozzle-assembly, and a device for modulating inlet velocity, as shown in Fig. 1.

The combustor is made of an optically accessible fused-silica combustor section (150 mm in diameter and 300 mm in length). The combustor exit is fully open without any restriction. The nozzle assembly consists of a center-bluff body (D=40 mm) whose end face is recessed by 20 mm from the dump plane and an annular mixing section where 45 deg swirl vanes are located. Figure 2 shows the schematic drawing of combustor and nozzle-assembly.

A siren type modulation device is used to impose the inlet-velocity fluctuation (V'). It consists of a variable-speed motor, a rotating plate, a preplenum, and a stator, and its modulation frequency range runs up to 400 Hz. The amount of inlet-velocity fluctuation (V'/V_{mean}) can be varied by controlling the relative amount of air flow through the modulating device. Natural gas is used as a fuel and is completely premixed with air at upstream of a choked inlet in order to avoid the fluctuation of equivalence ratio.

The velocity fluctuation in the annular mixing section is measured using a hot-wire anemometer (HWA) (TSI model 1210), which is located at 60 mm upstream of the dump plane. Chemiluminescence measurements are used to quantify the heat release response to inlet-velocity fluctuation. Photomultiplier tubes (PMTs) (Hamamatsu model 928A) are used to measure the global

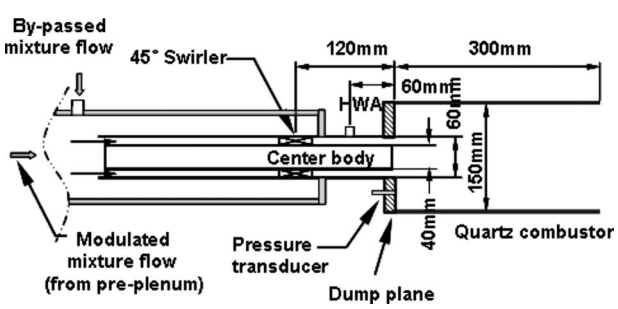

Fig. 2 Schematic drawing of the combustor and nozzle assembly

CH^* and OH^* chemiluminescence intensities from the whole flame, which are collected using a narrow band-pass ($\pm$5 nm) filter centered at 430 nm and 307 nm, respectively. The calculated values of $(CH^*)'/(CH^*)_{mean}$ or $(OH^*)'/(OH^*)_{mean}$ through the fast Fourier transform (FFT) are used as the estimates for the heat release fluctuations. The signals from HWA and PMTs are digitized and recorded by a digital data acquisition system at a sampling rate of 16,384 Hz. A total of 16,384 data points is taken during each test.

Also, an intensifed charge coupled device (ICCD) camera (Princeton Instruments model 576 G) equipped with a band-pass filter (λ=430$\pm$5 nm) is used to capture the CH^* chemiluminescence from the whole flame. Both time-averaged images over one period of inlet-velocity oscillation and phase-averaged images are taken for each condition. For all cases, 40 individual images are taken and averaged. For phase-averaged imaging, the camera is synchronized with the HWA signal, and 12 averaged images over one period of oscillation are taken with a phase interval of 30 deg. A three-point Abel deconvolution scheme is used to reconstruct the two-dimensional flame structure [16].

The operating conditions are summarized in Table 1. The experiments are conducted at room temperature and atmospheric-pressure conditions. Figure 3 shows the maximum and the minimum amplitude of inlet-velocity modulation that can be obtained over a range of frequency: The highest magnitude is found at the modulation frequency of 75 Hz and the lowest is found at around 230 Hz.

3 Results and Discussions

3.1 Global Characterization of Flame Response to Inlet-Velocity Fluctuation. Figure 4 shows typical time traces and frequency spectra of chemiluminescence intensity and inlet-velocity fluctuations measured for modulation frequency of 75 Hz, equivalence ratio of 0.7, and V'/V_{mean} of 0.07. Both CH^*- and OH^*-chemiluminescence intensities lag the inlet-velocity fluctuation, and the frequency spectra show that both inlet-velocity and chemiluminescence intensity fluctuations occur mainly at the

Table 1 Operating conditions

Mean velocity in the nozzle ($V_{mean,nozzle}$)	30 m/s
Equivalence ratio (ϕ)	0.65, 0.675, 0.7, 0.725, and 0.75
Inlet flow temperature	20°C
Modulation frequency (f)	75, 100, 140, 200, 250, and 280 (Hz)
Modulation amplitude (V'/V_{mean})	up to 21%
Reynolds number	35,600

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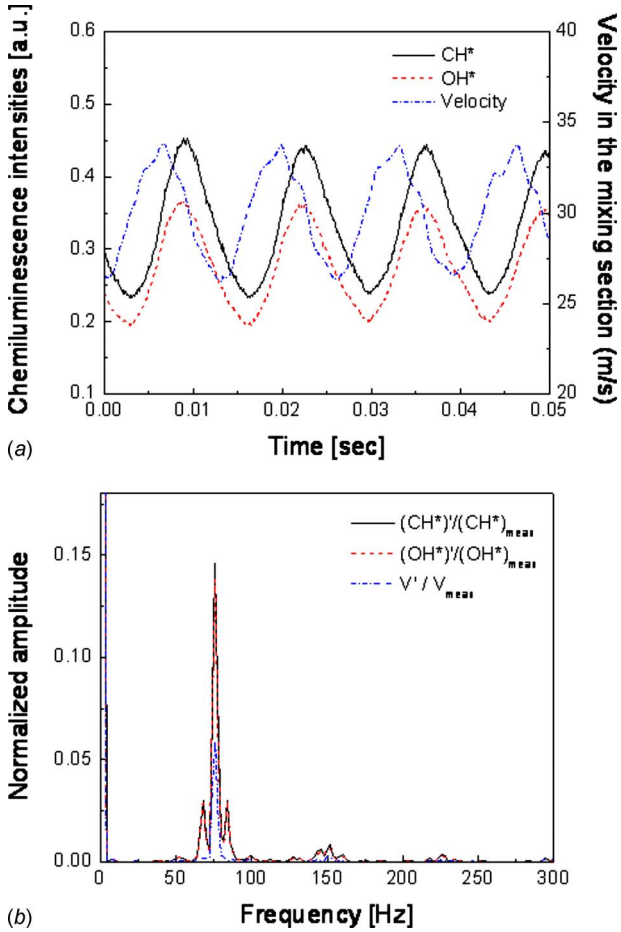

Fig. 4 (*a*) Time traces and (*b*) frequency spectra of CH* and OH* and inlet-velocity fluctuations ($V_{mean,nozzle}$=30 m/s, ϕ=0.7, f=75 Hz, and V'/V_{mean}=0.07)

modulation frequency with negligible amount of fluctuations at higher harmonic frequencies. Since the CH*- and OH*-chemiluminescence intensity fluctuations show similar behaviors both in amplitudes and in relative phases to the inlet-velocity perturbation over operating conditions listed in Table 1, only CH*-chemiluminescence results will be discussed hereafter.

Figure 5 shows averaged chemiluminescence images over 1 cycle of flow modulation at various modulation frequencies and equivalence ratios for a given mean velocity (V_{mean}=30 m/s). The modulation amplitude (V'/V_{mean}) was fixed at 0.07. In each

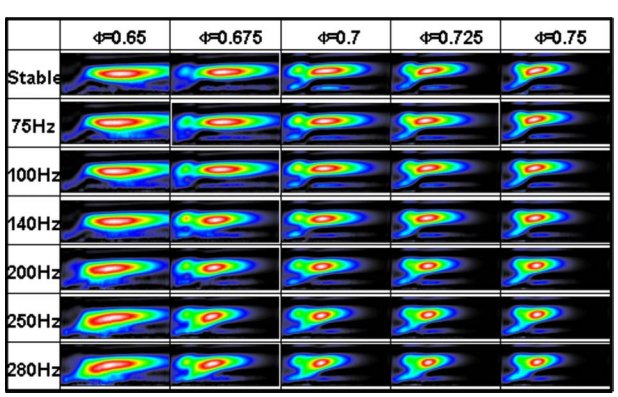

Fig. 5 Effects of modulation frequency and equivalence ratio on the flame structure ($V_{mean,nozzle}$=30 m/s and V'/V_{mean}=0.07)

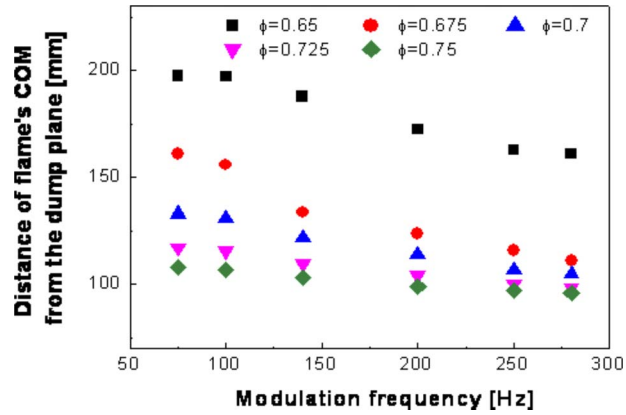

Fig. 6 Flame's COM location as a function of modulation frequency and equivalence ratio ($V_{mean,nozzle}$=30 m/s and V'/V_{mean}=0.07)

image, only the upper-half of the flame is shown because the deconvoluted images are axisymmetric. As mentioned in Sec. 2, the exposure time of the ICCD camera is set the same as the period of modulation so that each image represents an average of chemiluminescence images for one cycle of velocity fluctuation. These images show that the flame structure is affected by the modulation frequency as well as operating conditions (i.e., equivalence ratio): As the modulation frequency increases for a given equivalence ratio, the location for the maximum intensity region gets closer to the combustor inlet and the flame length decreases.

For more quantitative analysis of the change in flame structure, the location of center of mass (COM) from the dump plane is calculated for each image and plotted as a function of modulation frequency for various equivalence ratios, as shown in Fig. 6. It is evident that the distance of flame's COM from the dump plane decreases with increasing modulation frequency at a given equivalence ratio, and the trend is more obvious for the low equivalence ratio cases (say, ϕ=0.65 or 0.675) than the high equivalence ratio cases (say, ϕ=0.725 or 0.75). Since the location of COM can be interpreted as an effective flame location and/or the maximum heat release location, the results shown in Fig. 6 suggest that the actual convection time for the inlet-velocity perturbation to reach flame will change as the modulation frequency changes as well as inlet flow conditions. This will be further discussed in Secs. 3.2.

3.2 Transfer Function Measurement Results for a Fixed Amount of Inlet-Velocity Fluctuation. Since the effect of the magnitude of inlet-velocity fluctuation on transfer function measurement will be presented and discussed later, only cases for a fixed magnitude of input function (V'/V_{mean}=0.07) are presented in this section. Figure 7 shows the amplitude of CH* chemiluminescence fluctuation normalized by its mean value, which can be interpreted as the gain of transfer function (because the amplitude of velocity fluctuation is fixed) for different modulation frequencies and equivalence ratios. The gain of transfer function changes with respect to both inlet-velocity modulation frequency and inlet equivalence ratio. For a fixed equivalence ratio, the gain of transfer function decreases as the modulation frequency increases. Similar observations were made from analytical [14,15] and experimental works [7], where the global flame is found to behave as low-pass filter. Also, for a fixed modulation frequency, the gain increases as the equivalence ratio increases, indicating that the flame dynamics, which determines how flame responds to inlet-velocity fluctuation, depends on flame structure as well as on modulation frequency. As mentioned previously, analytical predictions have shown that the flame transfer function depends on flame angle [15] and flame length [14] for a given flame shape. In the present study, the flame structures such as flame length and the

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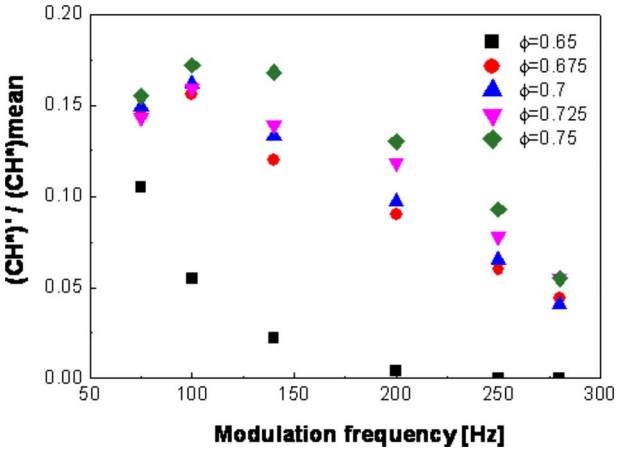

Fig. 7 Normalized CH* intensity fluctuation as a function of modulation frequency and equivalence ratio ($V_{mean,nozzle}$ =30 m/s and V'/V_{mean}=0.07)

existence of bright corner recirculation zone are changed with respect to the equivalence ratio, suggesting that the flame response to the same inlet-velocity fluctuation will change accordingly.

Figure 8 shows the phase-averaged CH* images taken for one cycle of inlet-velocity fluctuation for the equivalence ratio of 0.65 and modulation frequencies at 75 Hz and 280 Hz. The phase-averaging is synchronized with and referenced to the output signal from HWA. Low-frequency modulation (f=75 Hz) causes relatively larger amount of flame length change (as indicated by the variation in COM distance) as well as global chemiluminescence intensity (and hence heat release) than high-frequency modulation (f=280 Hz) does. For the 75 Hz case, the change in global heat release appears to be associated with the change in flame length even though they are not perfectly in phase, and the local heat release change occurs in the center part of the flame. However, for the 280 Hz case, the change in heat release in the center portion of the flame is minimal, and the flame front is more wrinkled, suggesting that the local flame stretch effect does not cause large variation in global heat release. For higher equivalence ratio case (ϕ=0.75) shown in Fig. 9, where the same amount of inlet-velocity modulation results in greater amount of fluctuations in heat release than ϕ=0.75 case, the heat release fluctuation seems to be associated with the change in flame length for both low- and high-frequency modulation cases. Also, the high-frequency modulation case (f=280 Hz) exhibits a periodic vortex-shedding process, which suggests that the flame response to inlet-velocity fluctuation may not be linear anymore at this condition, as suggested by Balachandran et al. [7].

Figure 10(a) shows the phase of transfer function between CH*-chemiluminescence intensity fluctuation, $(CH^*)'$, and V'. The absolute magnitude of the phase increases almost linearly as the modulation frequency increases for a given ϕ and decreases as ϕ increases for a fixed modulation frequency. Shown in Fig. 10(b) is the convection time delay (τ_{conv}) calculated from the phase. It shows that for a given ϕ, the higher the modulation frequency the shorter the delay between the heat release and inlet-velocity fluctuation, and for a given modulation frequency, the higher the equivalence ratio the shorter the delay. This is consistent with the effect of frequency and equivalence ratio on the flame's COM location shown in Fig. 6. This is because the closer the flame's COM location to the inlet of the combustor of the flame the shorter the delay.

From the data presented in Figs. 6 and 10, the effective convection velocity ($V_{eff}=L_{COM}/\tau_{conv}$) at which the inlet-velocity perturbation is convected in the combustor can be calculated. The results are shown in Fig. 11. The effective convection velocity is less than the mean inlet velocity to the combustor by a factor of 1.4–4 over the operating conditions and varies with respect to the modulation frequency as well as the equivalence ratio. For a given equivalence ratio, the effective convection velocity increases as the modulation frequency increases, seemingly suggesting that the effective convection velocity will increase as the flame becomes shorter. However, for a given modulation frequency, the effective convection velocity decreases as the equivalence ratio increases (as the flame length becomes shorter), suggesting that the effective convection velocity cannot be simply scaled with flame length but determined by the interaction of disturbance field with flame. One possible explanation for the reason why the effective convection velocity is not simply related to the flame length may have something to do with flame structures. Undisturbed flame

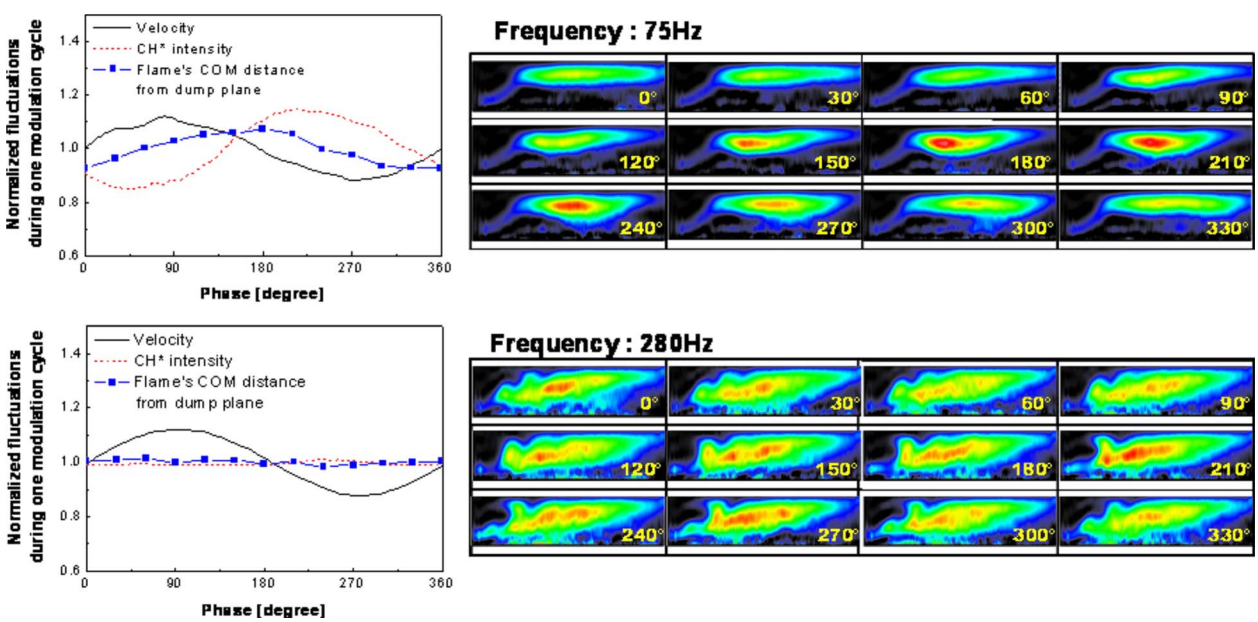

Fig. 8 Effects of modulation frequency on flame structure for one period of modulation ($V_{mean,nozzle}$=30 m/s, ϕ=0.65, and V'/V_{mean}=0.07)

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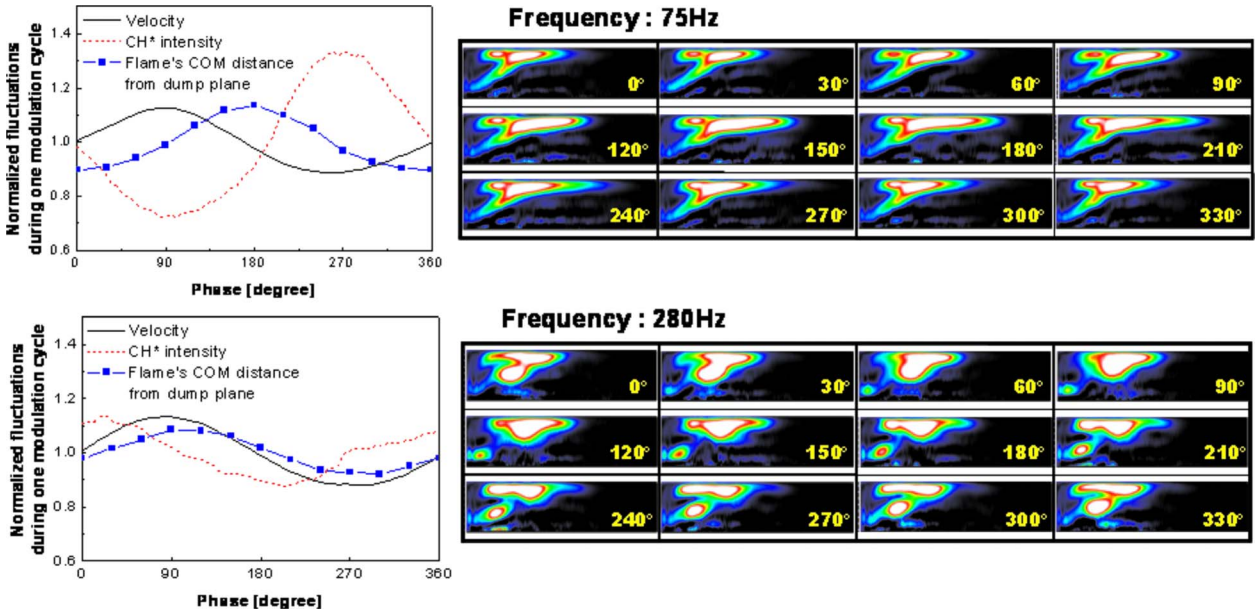

Fig. 9 Effects of modulation frequency on flame structure for one period of modulation ($V_{mean,nozzle}$=30 m/s, ϕ=0.75, and V'/V_{mean}=0.07)

image shows that for $\phi > 0.65$ cases, a bright heat release zone develops in the corner recirculation region, and this may cause the residence time of combustible mixture relatively longer for a given flame length than for $\phi = 0.65$ case.

Analytical studies [9,14,15] show that the dynamics of premixed flames submitted to harmonic flow perturbation with modulation frequency (f) can be characterized by the Strouhal number determined by

$$\mathrm{St} = \frac{L_{\mathrm{flame}}}{L_{\mathrm{conv}}} = \frac{L_{\mathrm{flame}}}{V_{\mathrm{eff}}/f}$$

which represents the ratio of flame length scale to the convective wavelength. Using the distance from the dump plane to the COM location shown in Fig. 6 as the flame length (L_{flame}) and the effective convection velocity shown in Fig. 11, the gain and phase of transfer function are plotted, as shown in Fig. 12. It shows that the flame length scale is always greater than the convection length scale (in other words $\mathrm{St} > 1$) for the present operating conditions. Also, these results show that the gain and phase of transfer function for different equivalence ratios and frequencies are well represented by the Strouhal number, strongly suggesting that the flame dynamics over different equivalence ratios and modulation frequencies could be generalized in terms of the relative length scale of the flame to the convection length scale.

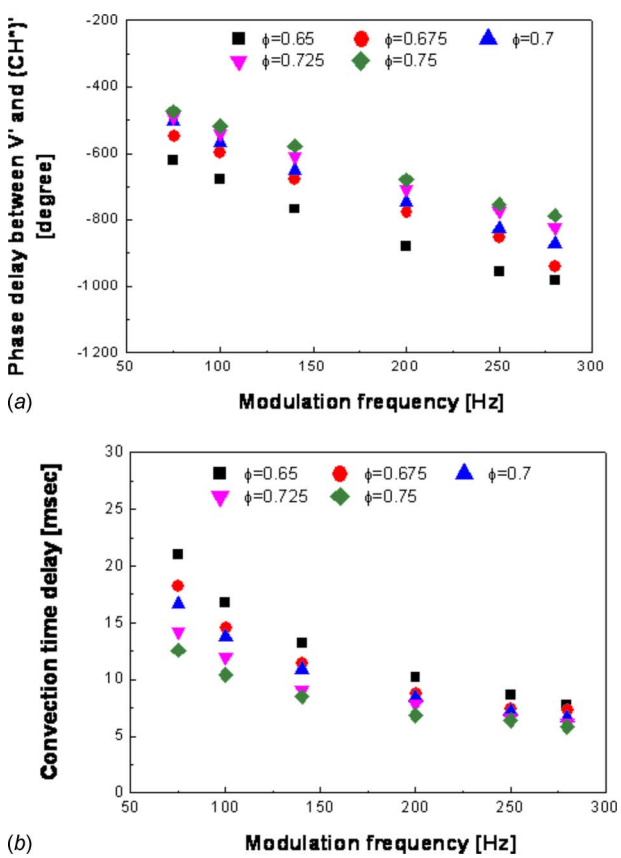

Fig. 10 (*a*) Phase of transfer function and (*b*) convection time delay as a function of modulation frequency and equivalence ratio ($V_{mean,nozzle}$=30 m/s and V'/V_{mean}=0.07)

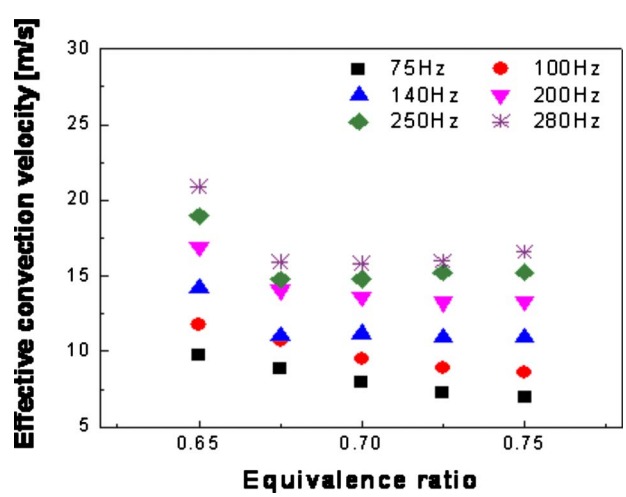

Fig. 11 Effective convection velocity as a function of modulation frequency and equivalence ratio ($V_{mean,nozzle}$=30 m/s and V'/V_{mean}=0.07)

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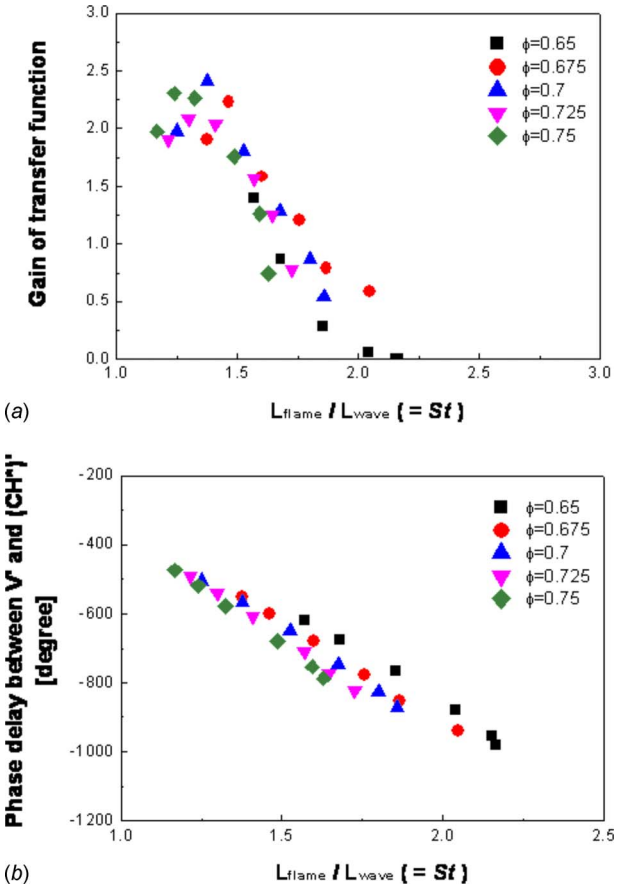

Fig. 12 (*a*) Gain and (*b*) phase of the transfer function versus Strouhal number ($V_{mean,nozzle}$=30 m/s and V'/V_{mean}=0.07)

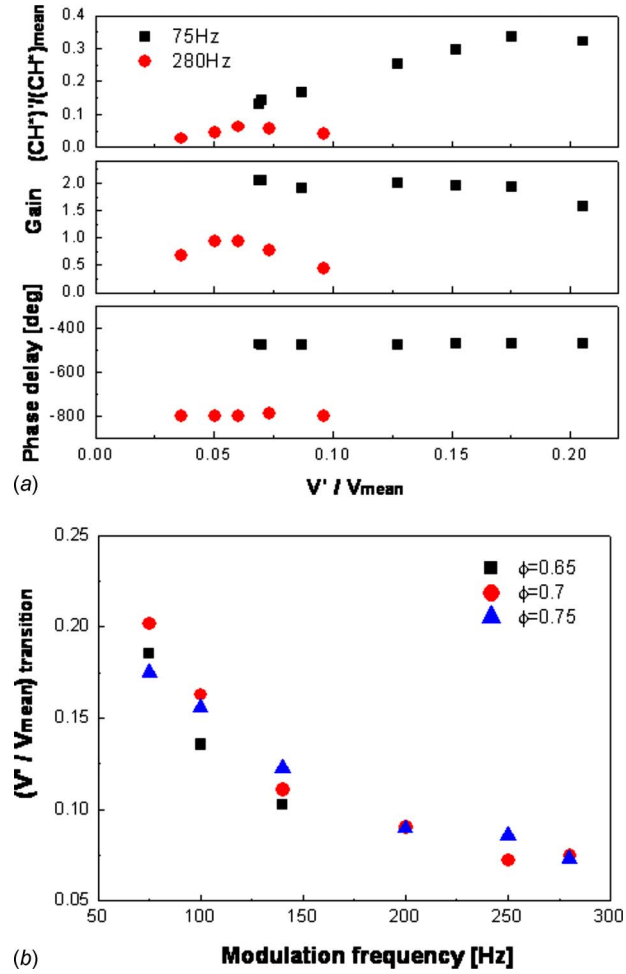

Fig. 13 (*a*) Normalized amplitude of CH* fluctuation, gain, and phase of the transfer function as a function of modulation amplitude for $V_{mean,nozzle}$=30 m/s and ϕ=0.75 and (*b*) magnitude of V'/V_{mean} at which the flame response changes from linear to nonlinear as a function of modulation frequency

3.3 Effect of the Magnitude of Inlet-Velocity Fluctuation on the Flame Transfer Function: Nonlinear Flame Response. Nonlinear response of flame to inlet-velocity perturbation is another important subject to understand the mechanism of combustion-driven oscillation. Figure 13(*a*) shows the dependence of transfer function on the inlet-velocity fluctuation amplitude at two different modulation frequencies (f=75 Hz and 280 Hz). Normalized CH*-chemiluminescence intensity increases and then start to saturate with increasing amplitude of inlet-velocity perturbation (V'/V_{mean}). The amplitude of inlet-velocity fluctuation at which the flame response becomes nonlinear for the 280 Hz modulation case is lower than that of the 75 Hz modulation case: For the latter the gain is almost constant up to V'/V_{mean} ~17%, beyond which it slightly decreases but for the former, the gain starts to decrease when V'/V_{mean} becomes over 7%. Previous analytical work [15] predicts that nonlinearity is enhanced with increasing Strouhal number. Similar observations are reported by some experimental works [7,8] where the inception of nonlinear flame response is often manifested by the occurrence of the rollup of the shear layer into vortices or flame liftoff, which reduces the flame area. In the present study, the nonlinear response of the flame is related to the vortex shedding from the shear layer, as shown in Fig. 9. Even though the gain decreases as V'/V_{mean} increases, the phase is almost constant over the entire range of inlet-velocity fluctuation. Figure 13(*b*) shows a plot of $(V'/V_{mean})_{transition}$ (the magnitude of V'/V_{mean} at which the flame response changes from linear to nonlinear) as a function of modulation frequency. As the modulation frequency increases, the nonlinearity occurs at lower V'/V_{mean}. However, for a given modulation frequency, $(V'/V_{mean})_{transition}$ does not appear to be affected much by flame structure.

4 Conclusions

The transfer function of premixed flames submitted to inlet-velocity perturbation is measured experimentally for turbulent flames running in practical gas turbine combustor. Flame structure is characterized under inlet-velocity perturbation, and its effect on transfer function is investigated.

Flame images show that the effective flame location gets closer to the combustor inlet with increasing modulation frequency at a given equivalence ratio. For a fixed equivalence ratio, the gain of the transfer function decreases and the absolute magnitude of the phase increases as the modulation frequency increases. And, their detailed relationships are affected by flame structure: For a fixed modulation frequency, the gain increases and the absolute magnitude of the phase decreases as the equivalence ratio increases. The gain and phase of the transfer function are well represented by the Strouhal number, strongly suggesting that the flame dynamics for different operating conditions and modulation frequencies could be generalized in terms of the relative length scale of the flame to the convection length scale of the inlet-velocity perturbation.

As the magnitude of inlet-velocity fluctuation increases, the gain starts to decrease but the phase is almost constant over the entire range of inlet-velocity fluctuation. This nonlinear flame response is characterized by a periodic vortex shedding from the shear layer. As the modulation frequency increases, the nonlinearity occurs at lower magnitude of inlet-velocity fluctuation. How-

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ever, for a given modulation frequency, the flame structure does not affect the magnitude of inlet-velocity fluctuation at which the nonlinearity starts.

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